

Exp No:16

SERIAL INTERRUPT USING UART (KEYBOARD INPUT) IN KEIL

AIM:

To write and simulate an Embedded C program using **serial interrupt** in 8051 microcontroller such that while a message is continuously transmitted, any key pressed from the keyboard generates an interrupt and is displayed through UART.

PROGRAM:

```
#include <reg51.h>
```

```
#include <stdio.h>
```

```
void serial_ISR(void) interrupt 4
```

```
{
```

```
char ch;
```

```
if (RI)
```

```
{
```

```
RI = 0; // Clear receive interrupt flag ch =
```

```
SBUF; // Read received character
```

```
printf("\n>>> INTERRUPT RECEIVED: ");
```

```
printf("%c\n", ch);
```

```
}
```

```
}
```

```
void delay(void)
```

```
{
```

```
    unsigned int i;
```

```
    for(i = 0; i < 50000; i++); // Delay for visibility }
```

```
void main(void)
```

```
{
```

```
    SCON = 0x50; // UART mode 1, 8-bit, REN enabled
```

```
    TMOD = 0x20; // Timer1 mode 2 (auto-reload) TH1 =
```

```
    0xFD; // Baud rate 9600
```

```
    TR1 = 1; // Start Timer1
```

```
    IE = 0x90; // EA = 1, ES = 1 (enable serial interrupt) TI
```

```
    = 1; // Enable printf
```

```
    while(1)
```

```
    {
```

```
        printf("Hello World\n");
```

```
        delay();
```

```
    }
```

```
}
```

The screenshot shows the Keil uVision IDE with the project 'EXP11.uvproj' open. The source file 'Exp_11.c' is displayed, showing the implementation of the delay function and the main loop. The main loop consists of a while(1) loop that prints 'Hello World\n' and calls the delay function. The UART1 window at the bottom right shows the output of the program, displaying 'Hello World' five times. The status bar at the bottom left indicates the limit is 2K and the file path is \\Examples\\HELLO\\Objects\\EXP11*.

```
20 void delay(void)
21 {
22     unsigned int i;
23     for(i = 0; i < 30000; i++); // Adjusted for simulator speed
24 }
25
26 void main(void)
27 {
28     SCON = 0x50; // Mode 1: 8-bit UART, REN enabled
29     TMOD = 0x20; // Timer 1, Mode 2: 8-bit auto-reload
30     TH1 = 0xFD; // 9600 Baud rate (for 11.0592 MHz crystal)
31     TR1 = 1; // Start Timer 1
32
33     IE = 0x90; // EA = 1 (Enable All), ES = 1 (Enable Serial Interrupt)
34     TI = 1; // Set TI flag to enable initial printf output
35
36     while(1)
37     {
38         printf("Hello World\n");
39         delay();
40     }
41 }
```

UART#1

```
Hello World
Hello World
Hello World
Hello World
Hello World
```